

A049 · Externally Validating Steered Disease Features in Single-Cell Foundation Models

Presenter: Mingxin Liu — Department of Biotechnology, Brown University

Authors: Mingxin Liu, Temidire Adesiji

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Single-cell foundation models learn rich representations of cellular states, but whether the disease-associated structure they capture relates to drug response in cancer cell lines remains unclear. We train a sparse autoencoder on scGPT activations and identify directions that separate tumors from normal cells in lung adenocarcinoma and invasive ductal breast carcinoma. We steer healthy cells along these axes, leading to monotonic shifts in their representations toward the tumor pole. We then test the induced shifts against expression changes measured in the Tahoe-100M cancer cell lines, and find no significant matches in either cohort after multiple hypothesis testing correction. scGPT thus carries steerable tumor directions that do not reproduce measured drug responses, though it is uncertain if the model or the cross-context comparison is responsible for this result.